3DGS-Matcher

Color-aware registration of independently built 3D Gaussian Splatting maps

Geometry-only matching: 6% → ours: 85% — training-free, on CPU

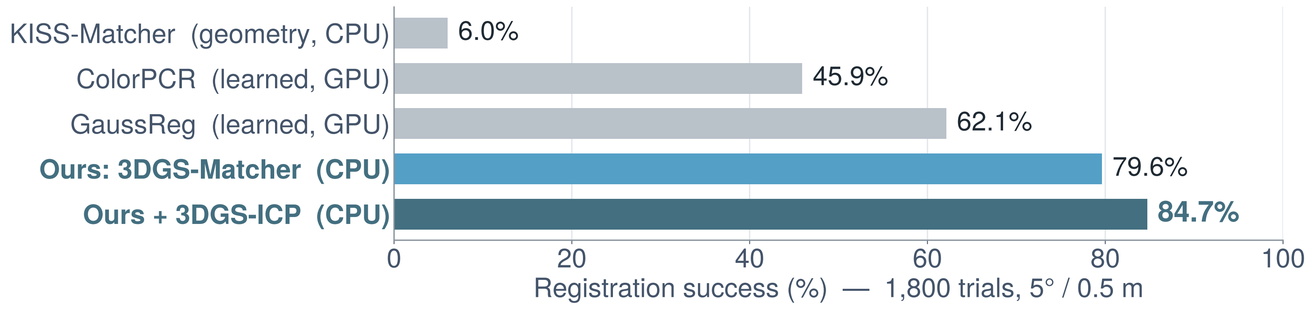
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The result in one glance

84.7%

success registering independently trained 3DGS maps

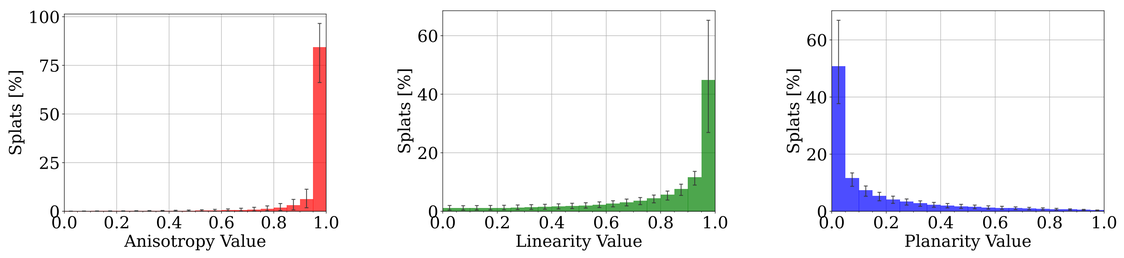
9 Mip-NeRF 360 scenes · 1,800 trials · 5° / 0.5 m



1 Why geometry fails for splats

Splats are fit to render, not to describe surfaces. Over 121 maps, their shapes pile up at the extremes:

84% anisotropy > 0.95 **45%** linearity > 0.95 **51%** planarity < 0.05

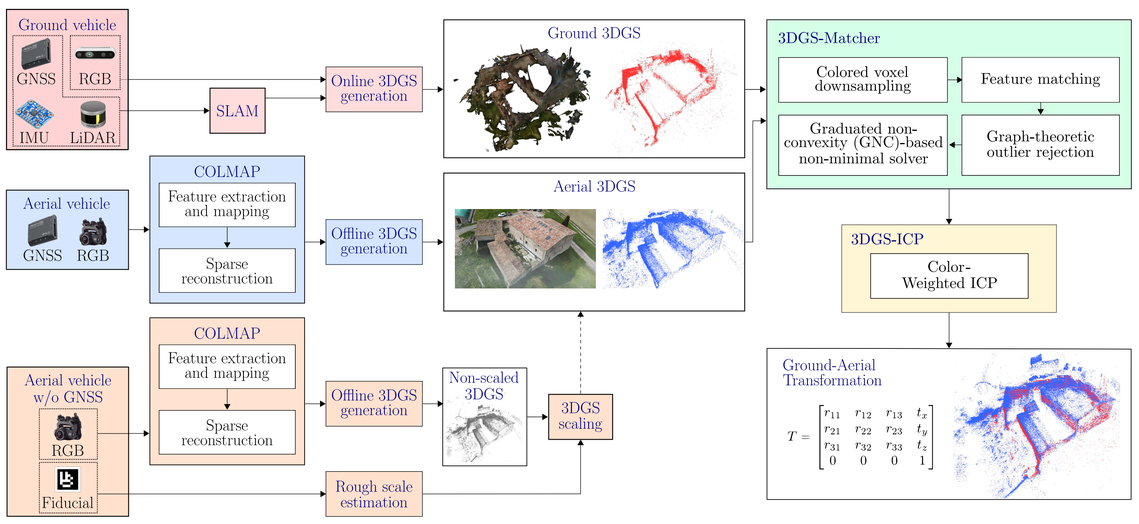


Takeaway

Geometry-only matching (KISS-Matcher): **6%**.

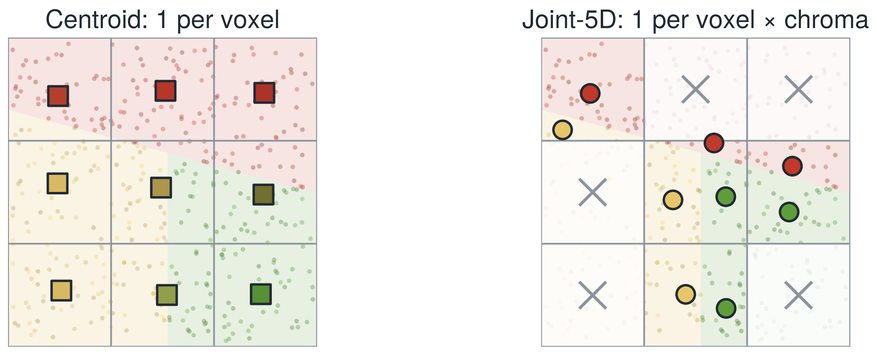
2 Two robots, no shared frame

Aerial and ground robots map independently; we align the maps from the Gaussians alone.

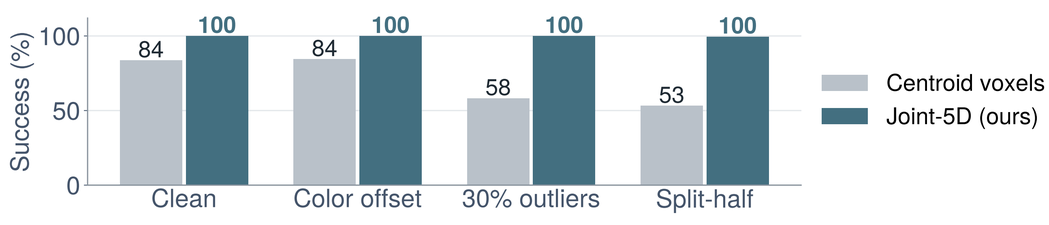


3 Coarse: match color, sample where color changes

ColorPFH: FPFH on CIELAB color differences. **Joint-5D:** one splat per voxel × chroma bin; single-color voxels dropped.



blended, made-up colors — real splats on color edges



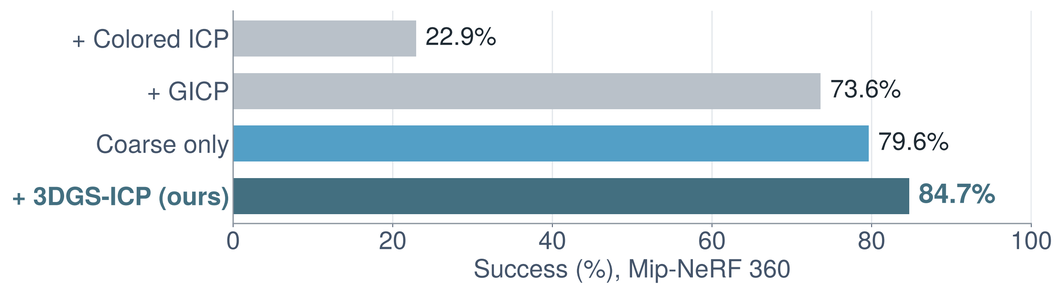
Takeaway

Same voxel size, better keypoints: **45%** → **80%**.

4 Refine: color-weighted ICP

GICP stays geometric; color only weights each correspondence:

$$w_{ik} = \epsilon + (1-\epsilon)\exp\left(-\frac{\|\mathbf{c}_i - \mathbf{c}_k\|^2}{2\sigma_c^2}\right)$$

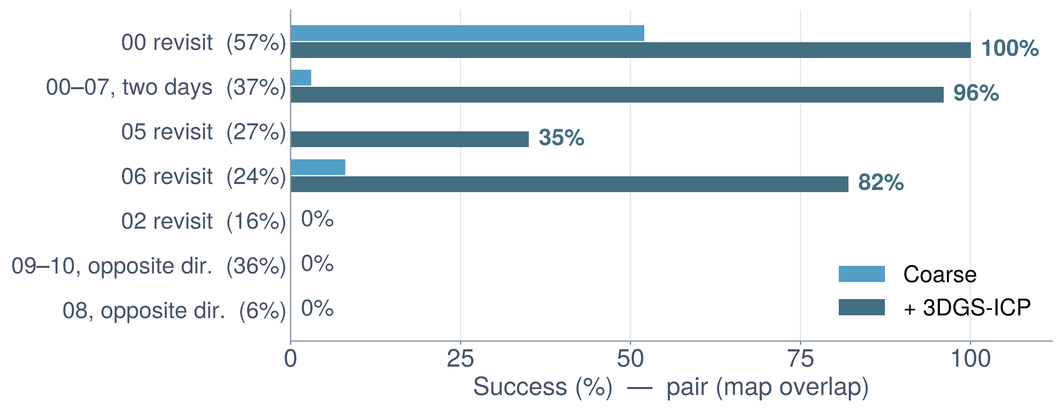


Takeaway

Color as a weight: **80%** → **85%**.

5 Multi-robot maps from KITTI

Each robot maps a disjoint stretch of a KITTI drive.

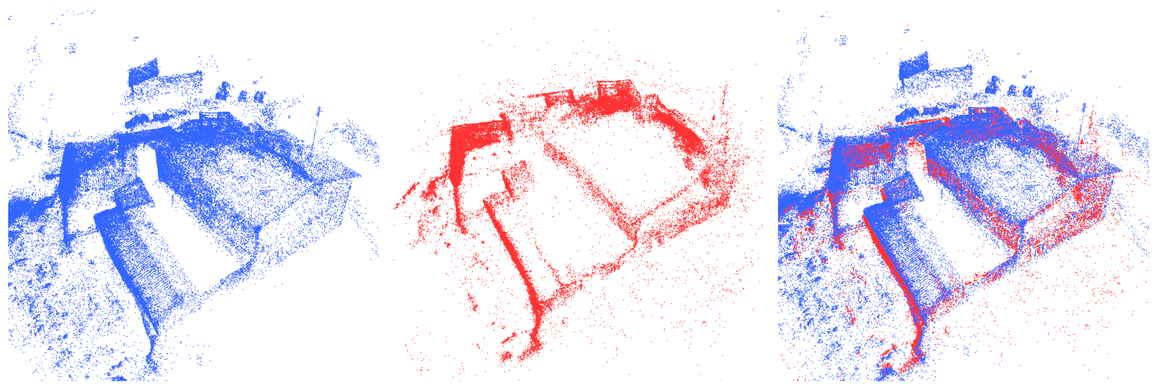


Takeaway

96% even across two days. KISS-Matcher: 0% on every pair.

6 Real aerial–ground maps

Drone vs. handheld RGB–LiDAR maps of a farmhouse.



85% of ground map within 0.5 m of aerial map

7 Limits & next step

Low color *stump*, *treehill*

Metric maps scale must be known

Clean input floaters filtered first

Next: Sim(3) registration, to align maps with no metric anchor

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References: Kerbl et al. (3DGS, 2023); Lim et al. (KISS-Matcher, 2025); Chang et al. (GaussReg, 2024); Mu et al. (ColorPCR, 2024); Barron et al. (Mip-NeRF 360, 2022); Geiger et al. (KITTI, 2012).